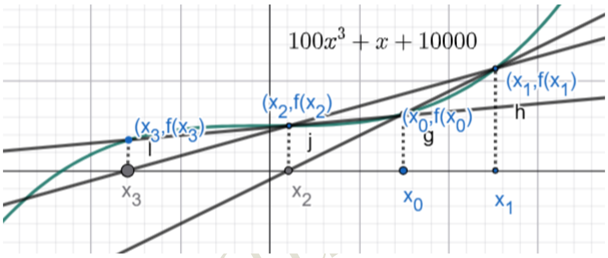
Name :	**Roll No.:**	**Program:**
Date: 2083/ /	**Computer No.:**	**Time:.........to ....**
Lab Status $\longrightarrow$	**Incomplete:**□	**Completed:**□
Checked by $\longrightarrow$	**Date:**	**Date:**

Lab 2: TO FIND A REAL ROOT OF A NON-LINEAR EQUATION BY SECANT METHOD USING PYTHON PROGRAMMING LANGUAGE

1 Theory



Secant method is a simple method to solve the non-linear equations under the category of open end method. It begins with two initial approximations x_0 and x_1 that do not necessarily bracket the root. The new approximation is point x_2, point of intersection of the secant line joining points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ of the curve $f(x)$ with x-axis.The latest two points x_0 and x_1 are used to find the next approximation. This keeps on going until the root is found with a given error. Here we approximate the root by the points of intersection of secant lines and x-axis in each iterations. This sequence converges to the true solution. This method is not guaranteed to convergence but is faster than bracketed methods. It is super-linearly convergent.

2 Algorithm Secant Method

(i) Start.

(ii) Define the non-linear equation $f(x)$ and error tolerance E.

(iii) Take two initial approximations $x = x_0$ and $x = x_1$.

(iv) If $f(x_0) = f(x_1)$, choose other approximations.

(v) Else compute the point $x_2 = \frac{x_0\ f(x_1) - x_1\ f(x_0)}{f(x_1) - f(x_0)}$.

(vi) If $|x_2 - x_1| < E$, then root $x = x_2$, print root. Go to step (ix).

(vii) Else, set $x_0 = x_1$, $x_1 = x_2$

(viii) Else go to step (iv).

(ix) Stop.

3 Pseudo-code

PRINT the title of the experiment
IMPORT necessary libraries
BEGIN IMPORT matplotlib.pyplot AS plt
IMPORT pandas AS pd
IMPORT numpy AS np

#Get equation from user
DISPLAY "Enter the equation in terms of x using python syntax: "
INPUT eqn

#Define function $f(x)$ using the input equation
FUNCTION $f(x)$
TRY
RETURN eval(eqn)
CATCH SyntaxError, NameError, TypeError AS e
DISPLAY "Invalid syntax: e
RETURN NULL
END TRY
END FUNCTION

Get initial guesses
DISPLAY "Enter two initial guesses: "
INPUT x_0 **AS FLOAT**
INPUT x_1 **AS FLOAT**

IF $f(x_0) - f(x_1) < 1e - 10$ **THEN**
DISPLAY "Division by zero! Choose new guesses!"
EXIT
ELSE
# Get tolerance and max iterations
DISPLAY "Enter tolerable error: "
INPUT E **AS FLOAT**
DISPLAY "Enter maximum number of iterations: "
INPUT N **AS INTEGER**

# Initialize variables
A = [ ] // Array to store root approximations
B = [ ] // Array to store iteration data
i = 1 // Iteration counter

#False Position Method Iterations
WHILE $i <= N$ **DO**
$x_2 = \frac{x_0\ f(x_1) - x_1\ f(x_0)}{f(x_1) - f(x_0)}$

Calculate error= $abs(x_2 - x_1)$
ADD x_2 **TO** A // Store approximation
ADD $x_0, x_1, x_2, f(x_0), f(x_1), f(x_2), error$ **TO** B
# Update values
IF $f(x_2) = 0$ **THEN**
break
Elif error$< E$
break
ELSE
$x_0 = x_1,\ x_1 = x_2$
END IF
$i = i + 1$

```
    END WHILE

# Check convergence status
    IF i > N THEN
        DISPLAY "Solution does not converge in N iterations"
    ELSE
# Convert iteration data to formatted table
        table= CREATE DataFrame FROM B
        WITH COLUMNS ['No. of iterations', 'x_0', 'x_1', 'x_2', 'f(x_0)', 'f(x_1)', 'f(x_2)', 'Error']
        DISPLAY table AS STRING WITHOUT INDEX

        root= x_2
        DISPLAY "The approximate solution is " + root + " in " + i + " iterations."
    END IF

# Visualization
    x = GENERATE_LINEAR_SPACE(x_0 - 5, x_1 + 5, 1000)
    y = CREATE_ZEROS_ARRAY(SAME_SIZE_AS(A))
    A = CONVERT_TO_ARRAY(A)

#Plot the function
    PLOT x VS f(x)
    ENABLE GRID
    DRAW_HORIZONTAL_LINE(0, COLOR =' red')
    DRAW_VERTICAL_LINE(0, COLOR =' red')

#Plot root approximations
    SCATTER_PLOT(A, f(A))

# Label iteration numbers
    FOR j = 0 TO LENGTH(A) - 1 DO
        PLACE_TEXT_AT(A[j], 0, TEXT = j + 1)
    END FOR

#Mark x-intercepts
    SCATTER_PLOT(A, y, MARKER =' x')

    DISPLAY PLOT
    END
```

4 Code:

5 Inputs and observations

S.N.	**Inputs**					**Output**		
	Equations	Initial Guesses		Error	Max. Iterations	Convergence Status	Approx. Solution	No. of iteration
1.	$e^x - x - 2 = 0$	1	2	1e-10	100			
		-2	1	1e-10	100			
		-6	6	1e-10	100			
		-6	-3	1e-10	100			
		-6	8	1e-10	100			
2.	$\cos x - \log x = 0$	1.2	1.4	1e-10	100			
		1.2	1.4	1e-10	100			
		-1	5	1e-10	100			
		-20	20	1e-10	100			
		-5	0	1e-10	100			
3.	$x - e^{-x} = 0$	0	-1	1e-10	100			
		0.5	0.6	1e-10	100			
		-1	1	1e-10	100			
		-15	15	1e-10	100			
		-10	10	1e-10	100			
4.	$e^x + \sin(x) - 9 = 0$	1	2	1e-10	100			
		2	2.1	1e-10	100			
		0	4	1e-10	100			
		9	10	1e-10	100			
		19	20	1e-10	100			

6 Graph:

7 Findings

(i)

(ii)

(iii)

(iv)

(v)

8 Conclusion: